

CS202: Programming Systems

Week 1: Object-oriented concept

10/2023

Object-oriented concepts

- ❑ Learning OO concepts is not accomplished by learning a specific development method or a set of tools.
- ❑ But, it is a way of thinking.

Object-oriented concepts (cont)

For examples:

- Many people are introduced to OO concepts via one of these development methods or tools.
 - ➔ Many C programmers were first introduced to object orientation by migrating directly to C++, before they were even remotely exposed to OO concepts.
 - ➔ Some software professionals were first introduced to object orientation by presentations that included object models using UML

Problems!!!

- ❑ Learning a programming language is an important step, but it is much more important to learn OO concepts first.
 - Developers who claim to be C++ programmers are simply C programmers using C++ compilers.
 - Learning UML before OO concepts is similar to learning how to read an electrical diagram without first knowing anything about electricity.

Even worse!!!

- A programmer can use just enough OO features to make a program incomprehensible to OO and non-OO programmers alike.

OO concepts

It is very important that while you're on the road to OO development, you first learn the fundamental **OO concepts.**

What is an object?

- ❑ For example: when you look at a person, you see the person as an object.
- ❑ An **object** is defined by two terms: **attributes** and **behaviors**.

An example: a person

- A person has attributes: eye color, age, height...
- A person also has behaviors: walking, talking, breathing, and so on.

**An *object* is an entity
that contains *both* data and behaviors**

Procedural vs. OO Programming

**An *object* is an entity
that contains *both* data and behaviors**

- In procedural programming:
 - Code is placed into totally distinct functions or procedures.
 - Data is placed into separate structures, and is manipulated by these functions or procedures.

Procedural vs. OO Programming (cont)

- ❑ In OO programming: the attributes and behaviors are contained within a single object
- ❑ In procedural programming: the attributes and behaviors are normally separated.

Why do we change from procedural to OO programming?



Why do we change from procedural to OO programming?

- In procedural programming:
 - Data is separated from the procedures.
 - Sometimes it is global → easy to modify data that is outside your scope
 - This means that access to data is **uncontrolled** and **unpredictable**.
 - Having no control over the data → testing and debugging are much more difficult.

Why do we change from procedural to OO programming?

- ❑ **Objects** solve these problems by combining data and behaviors into a **complete package**.
- ❑ In a proper OO design: there is no global data.

Objects (again!)

- ❑ Objects do contain:
 - Integers, and strings... → **attributes**.
 - Methods (i.e. functions) → **behaviors**.
- ❑ In an object, methods are used to operate on the data.

You can control access to members of an object (both attributes and methods).

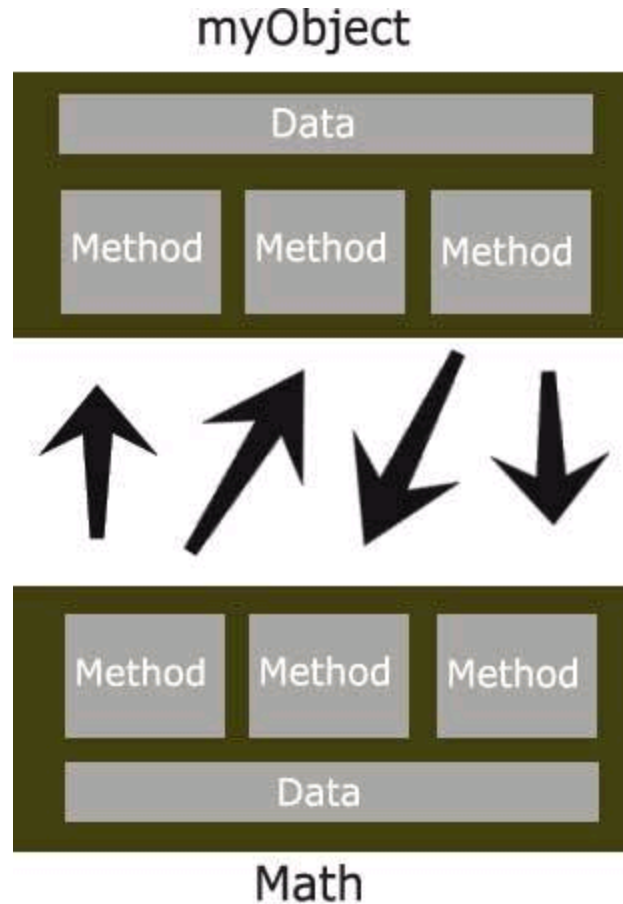
OO terminology

- ❑ Data is referred to as **attributes**.
- ❑ Functions are referred to as **methods**.
- ❑ Restricting access to certain attributes and/or methods is called ***data hiding***.

Encapsulation

- ❑ Combining the data and methods in the same entity.

Object-object communication



Moving from Procedural to Object-Oriented Development

- ❑ Procedural programming separates the data of the program from the operations.
- ❑ Example: if you want to send information across a network, only the relevant data is sent.
- ➔ handshaking agreement must be in place between the client and server to transmit the data.

Moving from Procedural to Object-Oriented Development

- In OO programming, when an object is transported across a network, the entire object, including the data and behaviors, goes with it.

What is an object (again!)?

- ❑ Objects are the building blocks of an OO program.
- ❑ A program that uses OO technology is basically **a collection of objects.**

Example – object data

- ❑ Let's consider that a corporate system contains objects that represent employees of that company.
- ❑ Employee attributes: ID, address date of birth, gender, phone number, and so on.
 - ➔ The attributes contain the information that differentiates between the various objects

Example – object behaviors

- ❑ The behaviors of an object are what the object can do
- ❑ In OO programming, these behaviors are contained in methods.
- ❑ You invoke a method by sending a message to it.

Exercise

- ❑ Define the attributes and behaviors for the object *Date*.
- ❑ Define the attributes and behaviors for the object *Shop* and *Items* in the shop .